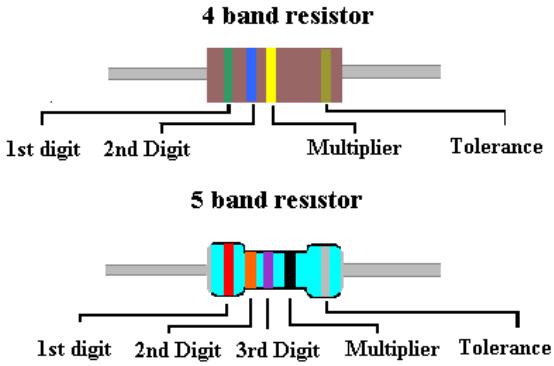
Chapter 44 – More on Electric Circuits
Calculus Physics II

Determining the value of a Resistor – Resistors are manufactured with color bands, this color code indicates the value of the resistance in Ohms and the uncertainty in the value of the given resistance. By looking at the different colors and determining their corresponding number value, the value of the resistor can be determined. The first two colors on the resistor gives the value of the first two digits of the resistance, and the third color indicates the power for the scientific notation (raised to the power of 10). The fourth digit is the uncertainty in the value given by the manufacture.

$$1^{st}\ 2^{nd} \times 10^{\text{3th digit}} \pm 4^{th}$$

Color code on resistors

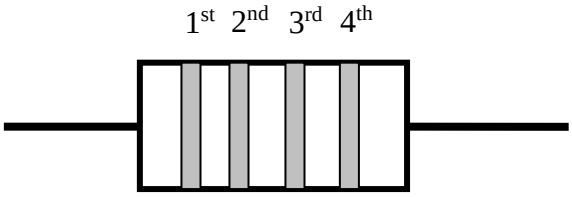
"Black	0	Black
Bears	1	Brown
Raid	2	Red
Our	3	Orange
Young	4	Yellow
Gardens	5	Green
But	6	Blue
Violet	7	Violet
Goats	8	Grey
Won't"	9	White



A way to remember the colors and values: "Black Bears Raid Our Young Gardens But Violet Goats Won't"

Uncertainty in the values

No band	20%
Silver	10%
Gold	5%
Pink	1%



$$1^{st}\ 2^{nd} \times 10^{\text{3th digit}} \pm 4^{th}$$

1st digit of value 2nd digit of value 3rd digit is number of zeros after second digit

4th digit is tolerance

Example of color on the resistor – if a resistor has color bands of;

a) Brown (1), red (2), red (2), gold (5%) $= 12 \times 10^2\ \Omega \pm 5\%$ or $1{,}200\ \Omega \pm 60\ \Omega$
b) Green (5), blue (6), black (0), silver (10%) $= 56 \times 10^0\ \Omega \pm 10\%$ or $56\ \Omega \pm 6\ \Omega$

Color of wires for wiring – in general dc and ac currents has different color wire:

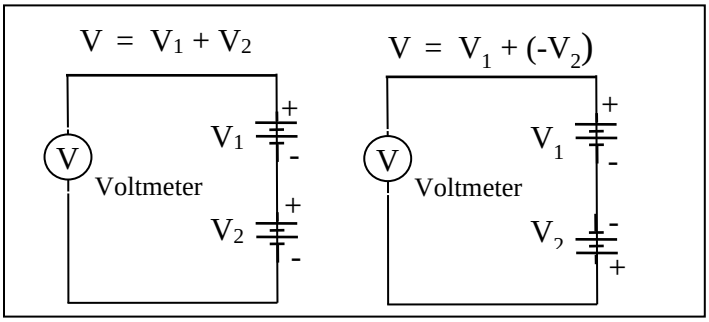
For dc – Red wire is positive. Black wire is negative or ground.

For ac - Black wire is Hot or high.
White wire is return wire or low (still dangerous!).
Green wire is grounding wire to device and to earth ground.

Dc Power Sources/Batteries in Series and Parallel - below are diagrams of dc power supplies wired in different configurations.

dc power sources wired in series for batteries wired with positive end connected to negative end.

$$V_{total} = \Sigma V_n = V_1 + V_2 + V_3 + \cdots$$

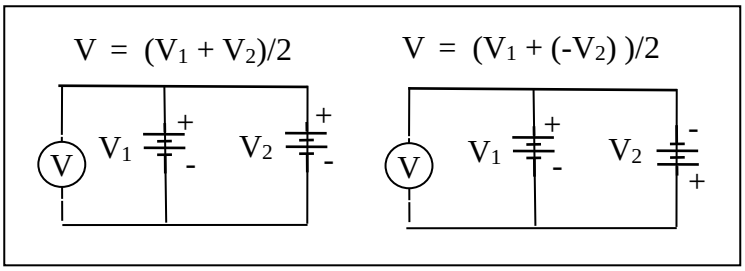


Note: It is not recommended to wire batteries so that the same pole is in contact.

- ✓ The total voltage across batteries in series that have their polarity in the same direction (positive to negative in same direction) is the sum of the voltages of each battery because the potentials added, each battery gives the charge more energy. Therefore, the combined power source supplies more energy per charge than the single battery.
- ✓ The total voltage across batteries in series that have their polarity in the opposite direction (positive to negative in opposite direction), potentials would subtract.

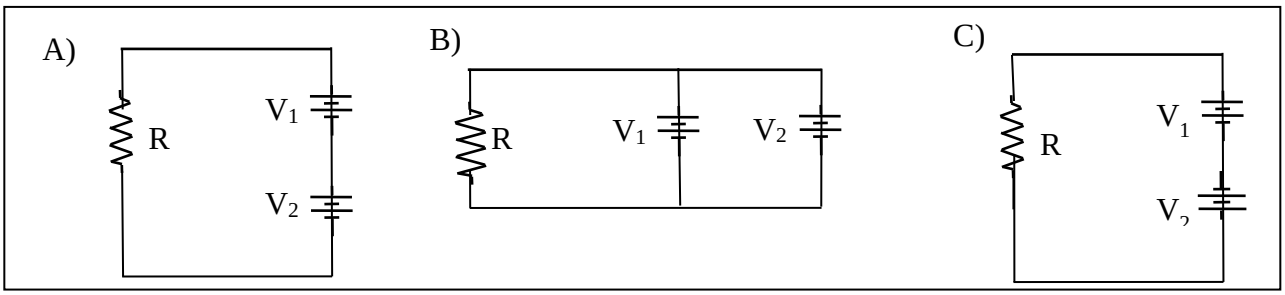
dc power sources wired in parallel for batteries wired with all positive poles at the same node.

$$V_{total} = \frac{\Sigma V}{n}$$



- ✓ The Voltage across batteries in parallel is their individual voltage because both power supplies have the same potential, this configuration can supply more charge to the circuit, therefore, more current can be supplied to the load. The voltage across the batteries is the average voltage of all the batteries in parallel.
- ✓ When batteries are in parallel but in opposite polarity, the plus end of one battery is on the same side as the minus end of the other, the voltages add but one is negative and the other positive, so they add to zero if they have the same voltage.

Example: If two batteries $V_1 = 50$ V and $V_2 = 20$ V are wired in the circuits below, how much current will flow through the resistors (R = 100 Ω)?



A) $V_{eq} = V_1 + V_2 = 50\text{ V} + 20\text{ V} = 70\text{ V}$ therefore

$I = V_{eq}/R = 70\text{V}/100\Omega = 0.7\text{ A}$

B) $V_{eq} = (V_1 + V_2)/2 = 35\text{ V}$ therefore

$I = V_{eq}/R = 35\text{V}/100\Omega = 0.35\text{ A}$

C) $V_{eq} = V_1 + (-V_2) = 50\text{ V} + (-20\text{ V}) = 30\text{ V}$ therefore

$I = V_{eq}/R = 30\text{V}/100\Omega = 0.3\text{ A}$

Internal Resistance of Power Sources – Sources of electrical power are not perfect; they do not deliver the total emf that they produce, there is always some loss due to internal resistance inside the power supply. This applies to battery and generators.

emf - is the "ideal" voltage of the source when no current is flowing, representing the maximum potential difference a source can produce.

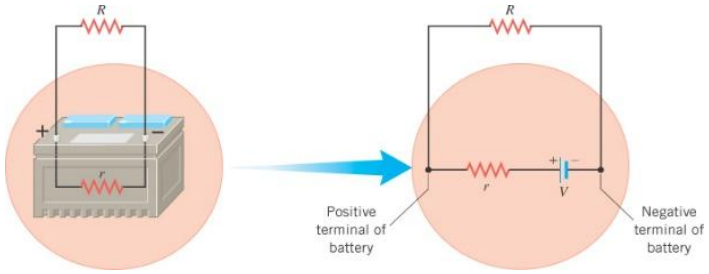
Terminal voltage - is the measured potential difference across the terminals of a device like a battery or generator when a current is flowing through it. Unlike the theoretical electromotive force (emf), terminal voltage is lower than the emf for a discharging battery due to the voltage drop across its internal resistance.

Terminal Voltage vs. Electromotive Force (emf) – For an ideal power source the emf is the maximum electric potential difference that exist between the terminals. In real power sources there is always internal loss, this leads to a potential difference across the power source that is less than the ideal emf. This loss increases with increased current flow, only when the current flow is zero can a real source produce a voltage difference that is its emf (emf = terminal voltage for I = 0 A).

- The emf is the ideal, maximum potential difference a battery can produce, terminal voltage is the voltage actually delivered to the external circuit.

Internal Resistance (r) - If there is internal resistance then the terminal voltage equals the emf minus the internal voltage drop (Ir) across the internal resistance where r is the internal resistance, and I is the current through the circuit and power supply.

$$V_{terminal} = V_{emf} - V_{internal} = V_{emf} - Ir$$



Electric generators - Internal Resistance is the Resistance/impedance of generators wire winding that produce the electrical power.

- If there is no internal resistance, the power supplies emf is equal to its terminal voltage

Example: A 9 V battery (V_{emf} = 9V) is wired in series with a switch and 10 Ω resistor

A) What does the voltmeter read across the battery when the switch is open, no current is flowing? For I = 0 A

$$V_{terminal} = V_{emf} - V_{internal} = emf = 9\ V$$

B) What is the internal resistance in the battery when the switch is closed a 0.7 A current flows and the voltmeter reads 7 V across the battery?

$V_{terminal} = emf - Ir$ Therefore $Ir = emf - V_{terminal}$

$r = (emf - V_{terminal})/I = (9\ V - 7\ V)/0.7A = 2.86\ \Omega$

Capacitors in Circuits - Series and Parallel – Capacitors are parallel plates that can accept a charge, this charge can move on or off the plates depending on the potential's polarity and how the capacitor is configured in the circuit.

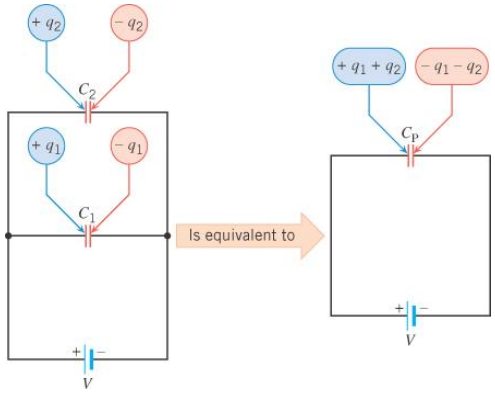
Capacitors in Parallel (C_p) – Equivalent Capacitor C_p - when capacitors are wired in parallel each capacitor "sees" the power source individually so the amount of charge each capacitor receives is as if it were in the circuit by itself. Therefore each capacitor effectively adds to the overall capacitance and the equivalent capacitance is the sum of all the capacitors capacitance in parallel.

For capacitors wire in paralell $V = V_1 = V_2 = V_3 = \cdots$

$C_p = C_1 + C_2 + C_3 + \ldots$ and $q_p = q_1 + q_2 + q_3 + \ldots = C_pV$

The total charge on all capacitors wired together in parallel q_p

$$q_p = C_pV$$



Example: If three 20 µF capacitors are wired in parallel and charged to 1,000 V

A) Determine the equivalent capacitance of the system and how much energy can it store at 100 V?

$C_p = C_1 + C_2 + C_3 = 20\ \mu F + 20\ \mu F + 20\ \mu F = 60\ \mu F$

Energy = $\frac{1}{2}C_pV^2 = \frac{1}{2}(60 \times 10^{-6}\ F)(1{,}000V)^2 = 30\ J$

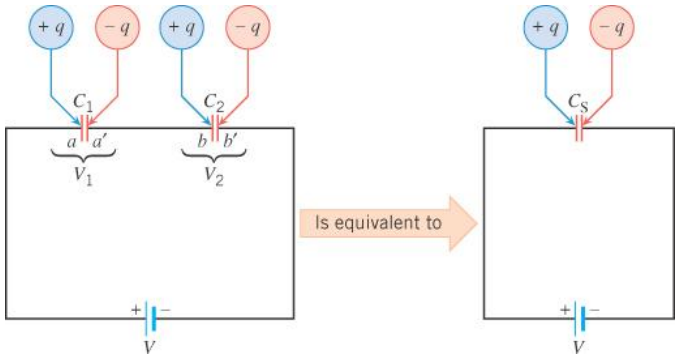
B) Determine the charge on each capacitor.

$$q_1 = C_1V_1 = 0.02\ C \qquad q_2 = C_2V_2 = 0.02\ C$$

$$q_{Total} = q_1 + q_2 = 0.04\ C$$

Capacitors in Series (C_s) - Equivalent Capacitance C_s when capacitors are wired in series the charge on each of the capacitors is the same, but the potential (V) is determined by the capacitance of the capacitor and the charge.

Looking at the wire between capacitor # 1 and capacitor # 2, the amount of charge must be the same on the bottom of C_1 and the top of C_2 because they are connected by a conducting wire, therefore the charge on top of C_1 must be the same as the charge on the bottom of C_2 which makes the charge on both capacitors equal.



The equivalent capacitance is given by

$$\boxed{\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots}$$

$$C_s = \frac{1}{\left[\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots\right]}$$

$$q_s = q_1 = q_2 = q_3 = \ldots = C_s V$$

The voltage across a capacitor number "n" wired in series is given by

$$V_n = \frac{q_s}{C_n} \quad so\ that \quad V_1 = \frac{q_s}{C_1} \quad V_1 = \frac{q_s}{C_2}$$

- ❖ All capacitors in series regardless of their capacitances contain charges of the same magnitude positive q and negative q on their plates.

Example: If three 20 µF capacitors are wired in series and charged to 1,000 V what is the equivalent capacitance of the system and how much energy can it store at 100 V?

$$C_s = \frac{1}{\frac{1}{20\ \mu F} + \frac{1}{20\ \mu F} + \frac{1}{20\ \mu F}} = 6.67\ \mu F$$

Energy = $\frac{1}{2}C_s V^2 = \frac{1}{2}(6.67 \times 10^{-6}\ F)(1{,}000V)^2 = 3.3\ J$

C) Determine the charge on each capacitor.

$$q_s = C_s V = 0.00667\ C$$

$$q_{Total} = q_s = q_1 = q_2 = 0.00667$$

D) Determine the voltage across each capacitor.

$$V_1 = \frac{q_s}{C_1} = 333\ V \qquad V_2 = \frac{q_s}{C_2} = 333\ V \qquad V_3 = \frac{q_s}{C_3} = 333\ V$$

- The sum of the voltage drops across the capacitor is equal to the total voltage in the circuit.
- Note that the voltage across each capacitor would be different for different capacitance values of the capacitors.

Energy Stored in the capacitors in a Circuit

Total Energy Series Circuit: Energy = ½ $C_s V^2$
Total Energy Parallel Circuit: Energy = ½ $C_p V^2$

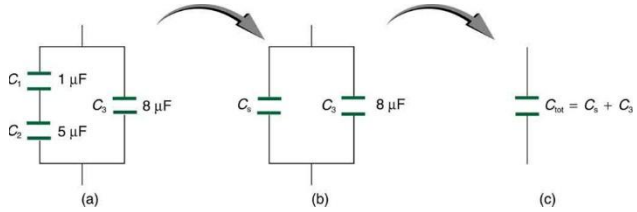
- More energy can be stored in capacitors in parallel than in series

Capacitors wired in series and parallel - Capacitors in series decrease the total capacitance, sharing voltage but having the same charge, while capacitors in parallel increase the total capacitance and each having different charges. Series capacitors adds voltage drop across each capacitor and, parallel capacitors adds charge, making combinations essential for specific capacitance and voltage needs in circuit design.

Example: Give three capacitors wired as shown, determine the equivalent capacitance of the circuit.

(a) Determine that capacitor 1 and 2 are in series

$$C_{1and2} = \frac{1}{\left[\frac{1}{C_1} + \frac{1}{C_2}\right]} = 0.83\ \mu F$$



(b) Draw the new circuit with the calculated equivalent capacitance.

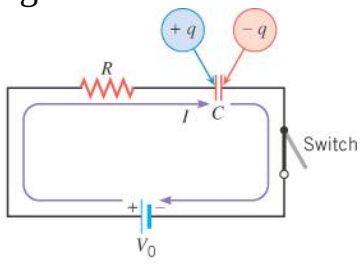
(c) Determining that capacitors are in parallel

$$C_{1and2and3} = C_{1and2} + C_3 = 8.83\ \mu F$$

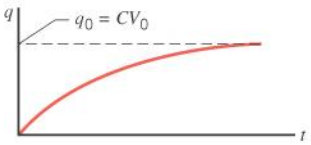
Resistor/Capacitor (RC) Circuits - Resistors and capacitors in the same circuit, the resistor act to slow down the charging or discharging process, larger resistance slower the charge/discharge.

Capacitor charging – When a capacitor that has no charge is place in a circuit that has a power supply and the switch is turned on, the charge flows onto the capacitor until it reaches its maximum limit of charge that is determined by the capacitance of the capacitor and the voltage across the capacitor. The capacitors have equal positive charge on one plate and equal negative charge on the other. This charging is not linear but starts out large (a lot of current at first) and tapers off over time. Charging a capacitor is a natural log function and is given as

$q(t) = q_o[1 - e^{-t/\tau}]$ $t = - RC\ln[(q_o-q)/q_o]$



(a)



(b)

q – charge at time t
q_o – maximum charge $q_o = CV$
e – Natural logarithm e = 2.718
C – capacitance
R – resistance
t – time of charging
τ – time constant = RC

Note: q(t) means that the charge depends on time not multiplied by time!

In electrical circuits the common quality used instead of charge is the potential across the capacitor. The voltage across a chagrining capacitor is given as a function of time V(t) and the amount of time to reach a given voltage (t).

$V(t) = V_o[1 - e^{-t/\tau}]$ $t = - RC\ln[(V_o-V)/V_o]$

Where V is the voltage at time t and V_o is the maximum changing voltage (power supply)

Time Constant τ (Tau) - is the amount of time required for the capacitor to accumulate 63.2 % of its maximum charge. The time constant indicates how fast a capacitor will charge or discharge. The time constant depends on the amount of resistance in the circuit and the size of the capacitor (the amount of capacitance in farads). The time constant is calculated using

$$\tau = RC$$

- The large the time constant the longer charging and discharging times

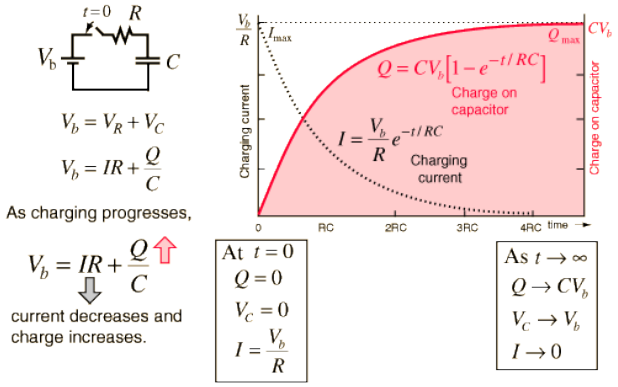
Limits of charging - When a capacitor charges, the current initially is large but gets smaller as time progresses because the greater charge on the capacitor makes it harder to move more charge to the plate and after some time the current approaches zero.

Using $V(t) = V_o[1 - e^{-t/\tau}]$

Knowing that $e^0 = 1$ then at t = 0 V = 0
Knowing that $e^{-\infty} = 1/\infty = 0$ then at $t = \infty$ $V = V_o$
Knowing that $e^{-1} = 0.37$ then when $t = \tau$ the capacitor is 1-0.37 = 0.63 or 63% charged

As the current goes to zero the voltage across the resistor goes to zero leaving the capacitors voltage equal to the batteries.

The maximum Charge $q_o = CV_o$. Where C is the capacitance of the capacitor and V is the maximum voltage on the capacitor (power supply).

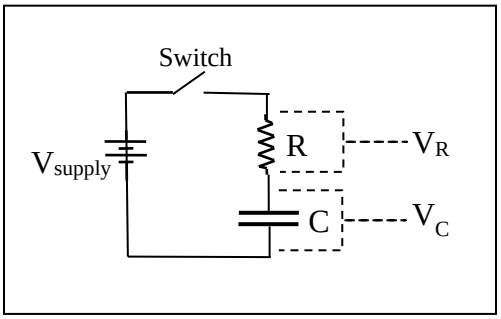


Charging Current in the RC Circuit - For a capacitor that is being charged, the current falls off rapidly over time and is given for a RC circuit with a voltage V and Resistance R the current falls off as

$$I(t) = I_o\, e^{-t/\tau}$$

Where t is the time and τ (Tau) is the time constant $\tau = RC$.

Voltage in the RC series circuit - For the voltage across the capacitor in the above circuit, when the switch is first turned on there is no charge on the capacitor so there is no voltage across the capacitor, all the voltage from the supply is dropped across the resistor. As time goes by the capacitor begins to accumulate charge therefore it begins to acquire a voltage, this voltage takes away from the resistor's voltage so that the total voltage between the resistor and capacitor still equals the supply voltage. Once the capacitor is fully charge there is no more current (charge) flowing and there is no voltage drop across the resistor and all the voltage from the supply is across the capacitor.



$$V_{supply} = V_R + V_C = I(t)R + q(t)/C$$

Example: A 12V battery is in series with a switch, resistor and uncharged capacitor.

If R = 20,000 Ω, V = 12 V, C = 100 μF

A) What values do the charge, current and voltage across the capacitor t = 2 seconds after the switch is closed and the capacitor begin to charge?

$\tau = RC = (20{,}000\Omega)(100 \times 10^{-6}\ F) = 2\ s$

$q_o = CV = (100 \times 10^{-6}\ F)(12V) = 1.2 \times 10^{-3}\ C$

Charge after 2 s

$$q(t) = q_o[1 - e^{-t/(RC)}] = (1.2 \times 10^{-3}\ C)[1 - e^{-2s/(2s)}] = (1.2 \times 10^{-3}\ C)[1 - 0.368]$$

$$q(t) = 7.58 \times 10^{-4}\ C$$

Current after 2 s

$$I_o = V/R = 12V/20000\Omega = 6 \times 10^{-4}\ A$$

$$I(t) = I_o\, e^{-t/\tau} = (6 \times 10^{-4}\ A)(e^{-2s/2s}) = (6 \times 10^{-4}\ A)(0.368)$$

$$I(t) = 2.2 \times 10^{-4}\ A$$

What is the volage across the resistor after 2 s

$$V = IR = (2.2 \times 10^{-4})(20{,}000\Omega) = 4.4\ V$$

Voltage across capacitor after 2 s and $V_{supply} = 12$ V

$$V(t) = V_o[1 - e^{-t/\tau}] = (12V)[1 - e^{-2s/2s}] = 7.6\ V$$

B) What values do the charge, current and voltage across the capacitor t= 20 seconds after the switch is closed and the capacitor begin to charge?

Charge after 20 s

$$q(t) = q_o[1 - e^{-t/(RC)}] = (1.2 \times 10^{-3}\ C)[1 - e^{-20s/(2s)}] = (1.2 \times 10^{-3}\ C)[1 - 4.5 \times 10^{-5}]$$

$$q(t) = 1.1999 \times 10^{-3}\ C \quad \text{(almost fully charged!)}$$

Current after 20 s

$$I_o = V/R = 12V/20000\Omega = 6 \times 10^{-4}\ A$$

$$I(t) = I_o\, e^{-t/\tau} = (6 \times 10^{-4}\ A)(e^{-20s/2s}) = (6 \times 10^{-4}\ A)(4.5 \times 10^{-5})$$

$$I(t) = 2.7 \times 10^{-8}\ A \quad \text{(almost no current!)}$$

What is the volage across the resistor after 20 s

$$V = IR = (2.7 \times 10^{-8})(20{,}000\Omega) = 5.4 \times 10^{-4}\ V \quad \text{almost zero!}$$

Voltage across capacitor after 20 s and V_{supply} = 12 V

$V(t) = V_o[1 - e^{-t/\tau}] = (12V)[1 - e^{-20s/2s}] = 11.999$ V
(almost all voltage drop across capacitor)

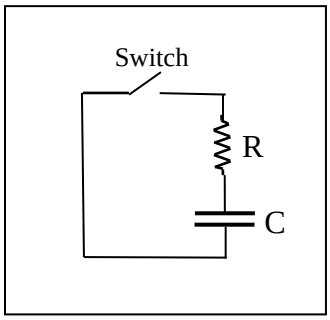
C) How long does it take to put half the charge on the capacitor? $q = \frac{1}{2}q_o$

$t = - RC\ln[(q_o-q)/q_o] = -(2s)\ln[(q_o-\frac{1}{2}q_o)/q_o] = -2(-0.693) = 1.36$ s

D) Determine the amount of time it would take to charge the capacitor to 10 V.

$t = - RC\ln[(V_o-V)/V_o] = - (2\ s)\ln[(12V-10)/12V] = 3.2$ s

Capacitor Discharge – When a charged capacitor discharges through a resistor as shown the capacitor loses its charge at a rate that depends on the resistance in the circuit and the capacitance of the capacitor.



$q(t) = q_oe^{-t/(RC)}$ $\quad$ $t = - RC\ln[q/q_o]$

Time constant $\tau = RC$ for discharge is the amount of time required for the capacitor have only 37 % of its equilibrium charge left on the plates or to losses 63 % of its charge.

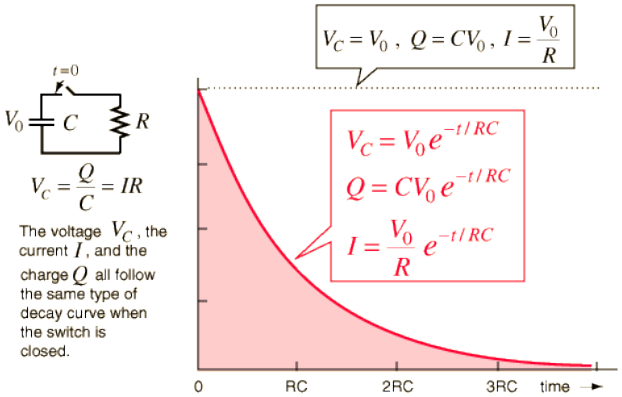
Note: The charged capacitor is the power source in the discharging circuit, there is no power supply.

The maximum Charge $q_o = CV$. Where C is the capacitance of the capacitor and V is the maximum voltage on the capacitor (power supply).

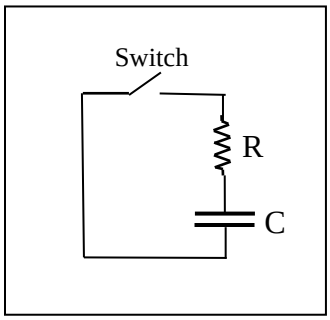
The voltage across a capacitor that is discharging is given by

$V(t) = V_oe^{-t/(RC)}$ $\quad$ $t = - RC\ln[V/V_o]$

Where V is the voltage at time t and V_o is the maximum changing voltage (power supply)



Example: A fully charged capacitor (V) is in series with a switch and resistor. (No battery, the power source is the charged capacitor!)



R = 20,000 Ω, V = 12 V, C = 100 μF

A) How much charge is still on the capacitor 2 seconds after the switch is closed and the capacitor begin to discharge?

$\tau = RC = (20{,}000\Omega)(100 \times 10^{-6}\ F) = 2\ s$

$q_o = CV = (100 \times 10^{-6}\ F)(12V) = 1.2 \times 10^{-3}\ C$

Charge after 2 s of discharge

$q(t) = q_o[e^{-t/(RC)}] = (1.2 \times 10^{-3}\ C)[e^{-2s/(2s)}] = (1.2 \times 10^{-3}\ C)[0.368]$

$q(t) = 4.4 \times 10^{-4}\ C$

Charge after 20 s of discharge

$q(t) = q_o[e^{-t/(RC)}] = (1.2 \times 10^{-3}\ C)[e^{-20s/(2s)}] = (1.2 \times 10^{-3}\ C)[\ 4.5 \times 10^{-5}]$

$q(t) = 5.4 \times 10^{-8}\ C$ (very little charge left on capacitor)

B) How much time does it take to discharge half the charge from the capacitor? $q = ½q_o$

$t = - RC\ln[q/q_o] = - (2s)[\ln(0.5)] = 1.39\ s$

C) What is the voltage across the capacitor after 3 s?

$V(t) = V_o e^{-t/(RC)} = (12V)\ e^{-30s/(2s)} = 2.7\ V$

D) How much time does it take the capacitor to reach a volage of 1 V?

$t = - RC\ln[V/V_o] = -(2\ s)\ln(1V/12V) = 5\ s$